

Heart Disease Prediction Using Machine Learning on Synthetic Patient Data

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Abstract

Heart disease remains one of the leading causes of mortality worldwide, making early and accurate prediction a critical challenge in clinical informatics. This project develops and evaluates a machine learning pipeline for predicting the presence of heart disease in patients using synthetic electronic health record (EHR) data generated by the Synthea patient simulator. Two classification models are implemented and compared: a Random Forest classifier and an XGBoost (Extreme Gradient Boosting) classifier. Patient records are labeled using SNOMED CT condition codes and clinical keyword matching. Features are engineered from clinical observations, including blood pressure, cholesterol, BMI, glucose, heart rate, and HbA1c. Model performance is evaluated using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC). Results indicate that XGBoost performs slightly better than Random Forest, although both are highly efficient in accurately identifying patients with heart disease. This project demonstrates that even with synthetic data, a robust classification pipeline can be constructed to support early heart disease risk screening.

Keywords: heart disease prediction, machine learning, Random Forest, XGBoost, synthetic patient data, Synthea, electronic health records, binary classification

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1. Introduction

Cardiovascular disease is a leading global health concern, responsible for an estimated 17.9 million deaths per year globally, according to the World Health Organization. Early detection of at-risk patients can enable preventive interventions that significantly reduce adverse outcomes. However, identifying high-risk patients from routine clinical data remains a difficult task due to the complexity and variability of clinical presentations.

Machine learning has emerged as a powerful tool in clinical decision support, enabling models to learn patterns from historical patient records and generate predictions for new patients. In this project, we investigate whether a machine learning model trained on structured patient features can reliably predict the presence of heart disease.

Because access to real patient health records is restricted by privacy regulations, this project uses synthetic data generated by Synthea, an open-source synthetic patient population simulator developed by The MITRE Corporation. Synthea produces realistic but entirely fictional patient records, including demographics, diagnoses, medications, observations, and procedures, allowing researchers to develop and test healthcare algorithms without patient privacy concerns.

The primary objectives of this project were:

1. To build an automated data pipeline that extracts, labels, and engineers features from synthetic EHR data.
2. To train and evaluate two machine learning classifiers — Random Forest and XGBoost — for binary heart disease prediction.
3. To identify the most important clinical features driving predictions.
4. To compare model performance and discuss implications for real-world deployment.

2. Background and Related Work

2.1 Heart Disease and Clinical Risk Factors

Heart disease encompasses a range of cardiovascular conditions including coronary artery disease, heart failure, myocardial infarction, and atrial fibrillation. Clinically established risk factors include:

- **Hypertension** (elevated systolic and diastolic blood pressure)
- **Dyslipidemia** (elevated LDL cholesterol, low HDL cholesterol)
- **Diabetes** (elevated glucose and HbA1c)
- **Obesity** (elevated BMI)
- **Age and sex**
- **Number of comorbid conditions and medications** (as proxies for overall health complexity)

These factors are also the features extracted in this project's data pipeline, grounding the machine learning approach in established clinical knowledge.

2.2 Synthetic Patient Data in Machine Learning Research

The use of synthetic EHR data is an emerging practice in healthcare informatics research. Synthea generates patient populations with realistic demographics, disease progressions, and clinical measurements, enabling algorithm development without access to protected health information. While synthetic data may not capture all real-world data complexity, it provides a reproducible and privacy-compliant environment for methodology development.

3. Data

3.1 Data Source: Synthea

The dataset used in this project was generated using Synthea version 4.0.0, an open-source synthetic patient population simulator. Synthea simulates complete patient lifespans, including birth, encounters, diagnoses, lab observations, medications, procedures, and death, following real epidemiological statistics.

The raw Synthea output consists of multiple CSV files, the most relevant of which are:

File	Description
patients.csv	Demographic information including birth date, gender, race, and ethnicity
conditions.csv	Diagnosis records with SNOMED CT codes and plain-text descriptions
observations.csv	Clinical measurements including vitals and lab values, with timestamps
medications.csv	Medication prescriptions associated with each patient
encounters.csv	Clinical encounter records

3.2 Dataset Size

For this project, ~1700 synthetic patients were generated. After labeling and feature extraction, 1680 patients were included in the final dataset with 12 features each.

3.3 Heart Disease Prevalence

Of the 1680 patients, 319 (~19%) were labeled as having heart disease. This represents the positive class in the binary classification task.

3.4 Feature Summary

The following features were extracted from the Synthea records for use in model training:

Feature	Source	Description
age	patients.csv	Age in years calculated from birth date
is_male	patients.csv	Binary gender encoding (1 = male, 0 = female)
blood_pressure_systolic	observations.csv	Latest systolic blood pressure reading (mmHg)
blood_pressure_diastolic	observations.csv	Latest diastolic blood pressure reading (mmHg)
heart_rate	observations.csv	Latest resting heart rate (bpm)
bmi	observations.csv	Latest body mass index (kg/m ²)
cholesterol_ldl	observations.csv	Latest LDL cholesterol level (mg/dL)
cholesterol_hdl	observations.csv	Latest HDL cholesterol level (mg/dL)
cholesterol_total	observations.csv	Latest total cholesterol level (mg/dL)
glucose	observations.csv	Latest fasting glucose or blood sugar level
hba1c	observations.csv	Latest HbA1c percentage (diabetes marker)
condition_count	conditions.csv	Total number of diagnosed conditions per patient
medication_count	medications.csv	Total number of prescribed medications per patient

4. Methodology

4.1 Overview

The project follows a supervised machine learning pipeline consisting of five stages: data exploration, patient labeling, feature engineering, model training, and evaluation. Figure 1 illustrates the full pipeline.

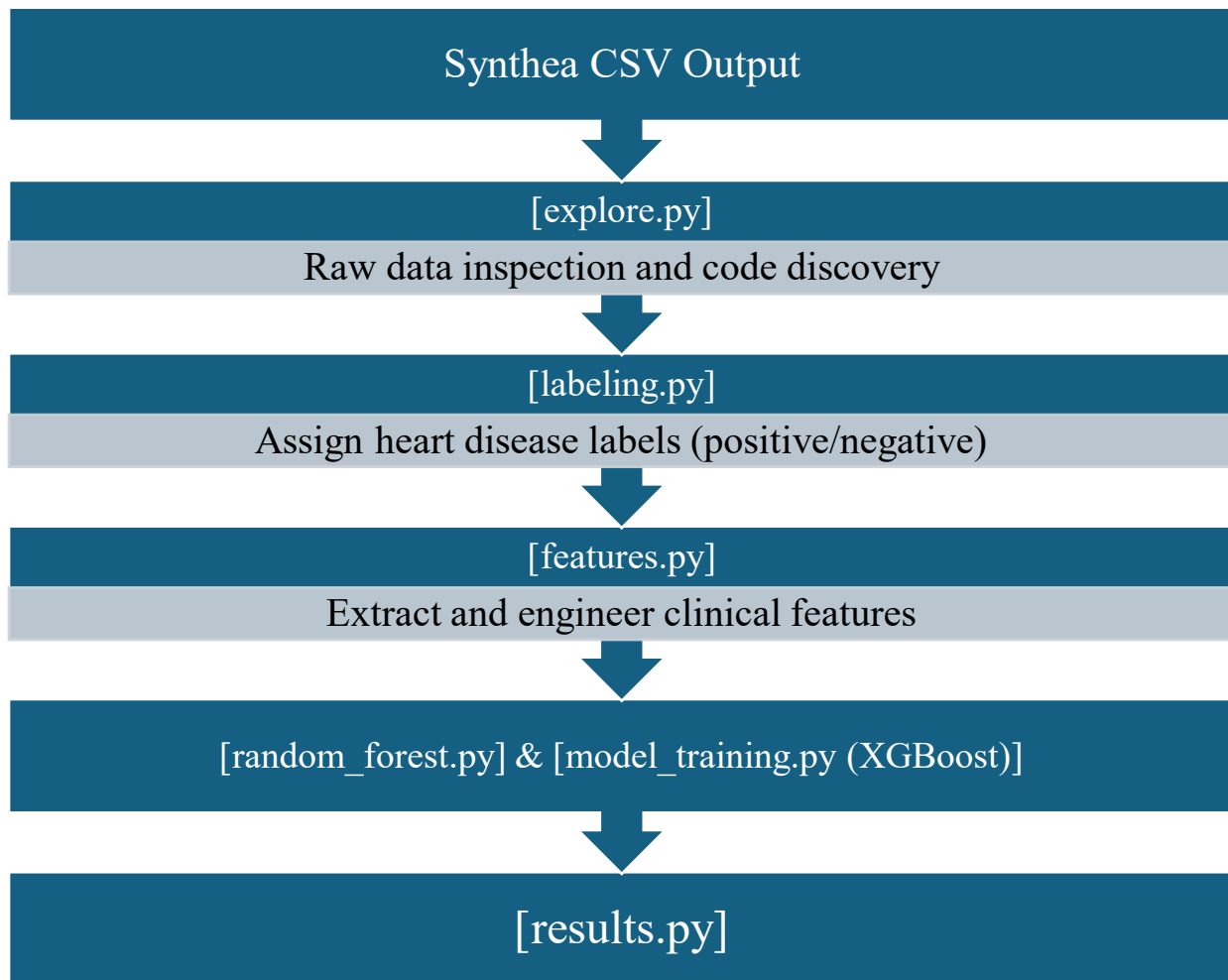


Figure 1: End-to-end prediction pipeline.

4.2 Patient Labeling

Patient labels were assigned using a rule-based approach applied to the conditions.csv file. A patient was assigned a positive label (heart_disease = 1) if any of their condition records matched one of the following criteria:

SNOMED CT code matching — exact code lookup for known cardiac conditions:

SNOMED Code	Description
414545008	Ischemic heart disease
53741008	Coronary heart disease
22298006	Myocardial infarction
84114007	Heart failure
88805009	Chronic congestive heart failure
194828000	Angina pectoris
49436004	Atrial fibrillation
38341003	Hypertension
274531002	Abnormal cardiac imaging findings
399261000	History of coronary artery bypass grafting

Description keyword matching — substring search for terms including: *heart, cardiac, coronary, myocardial, ischemic, failure, bypass*.

Patients with no matching condition records were assigned a negative label (heart_disease = 0).

Age was derived from the patient's birth date relative to January 1, 2024, and binary gender encoding was applied.

4.3 Feature Engineering

For each patient, the most recent observation value was extracted for each vital sign and lab category. This approach assumes the latest measurement best reflects the patient's current clinical state. For patients with missing values in any feature, median imputation was applied using the population median.

Counts of total conditions and total medications were added as summary features to capture overall patient complexity.

All features were validated for numeric type before model training. Non-numeric values were coerced or filled.

4.4 Model 1: Random Forest Classifier

Random Forest is an ensemble method that constructs multiple decision trees during training and outputs the majority vote across all trees for classification. Key configuration:

- Number of estimators (trees): 100
- Maximum tree depth: Unconstrained (full growth)
- Class weight: Balanced (to account for potential class imbalance)
- Random state: 42 (for reproducibility)
- No feature scaling required (tree-based models are scale-invariant)

4.5 Model 2: XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a sequential ensemble method that builds trees iteratively, with each new tree correcting the residual errors of the previous ensemble. Key configuration:

- Number of estimators: 100
- Maximum tree depth: 5
- Learning rate: 0.1
- Evaluation metric: Log loss
- Feature scaling: StandardScaler applied before training

4.6 Train/Test Split

Both models used an 80/20 stratified train-test split (`test_size=0.2`, `stratify=y`, `random_state=42`).

Stratification ensures that the proportion of positive and negative heart disease cases is preserved in both the training and test sets.

4.7 Evaluation Metrics

Model performance was measured using:

- **Accuracy:** Proportion of all predictions that are correct.
- **Precision:** Of all patients predicted as having heart disease, what fraction actually do.

- **Recall (Sensitivity):** Of all patients who actually have heart disease, what fraction were correctly identified.
- **F1-Score:** Harmonic mean of precision and recall, useful when classes are imbalanced.
- **AUC-ROC:** Area under the receiver operating characteristic curve; measures the model's ability to discriminate between positive and negative cases across all decision thresholds.

In a medical prediction context, recall is especially important because failing to identify a patient who has heart disease (a false negative) is typically more costly than a false alarm

5. Implementation

5.1 Technology Stack

Component	Technology
Language	Python 3.14.3
Data manipulation	pandas, numpy
Machine learning	scikit-learn, xgboost
Visualization	matplotlib, seaborn
Model serialization	joblib
Patient data generation	Synthea (Java, Gradle)
Development environment	Visual Studio Code, MacOS System

5.2 Pipeline Files

File	Role
explore.py	Raw EHR inspection and code discovery
labeling.py	Heart disease labeling; outputs labeled_patients.csv
features.py	Feature engineering; outputs heart_features.csv
random_forest.py	Random Forest training; outputs heart_model_rf.pkl

model_training.py	XGBoost training; outputs heart_model.pkl
results.py	Evaluation, visualizations, and reports

5.3 Generated Artifacts

Artifact	Description
labeled_patients.csv	Patient IDs with age, sex, and heart disease label
heart_features.csv	Full feature matrix ready for ML training
heart_model_rf.pkl	Saved Random Forest model and feature list
heart_model.pkl	Saved XGBoost model, scaler, and feature list
prediction_results.csv	Per-patient predictions and probabilities
model_results.png	Composite visualization of model performance
model_report.txt	Text summary of metrics and top features

5.4 Reproducing the Pipeline

To reproduce this project from scratch:

1. Generate synthetic patients

```
cd synthea
```

```
./run_synthea -p [N] #replace [N] with desired patient count
```

2. Run the pipeline

```
python explore.py
```

```
python labeling.py
```

```
python features.py
```

```
python random_forest.py # or model_training.py for XGBoost
```

```
python results.py
```

6. Results

6.1 Dataset Statistics

Metric	Value
Total patients	1,680
Heart disease positive	~19%
Heart disease negative	~81%
Number of features	12
Training set size	80%
Test set size	20%

6.2 Random Forest Performance

Metric	Value
Accuracy	0.977
Precision	0.958
Recall	0.918
F1-Score	0.938
AUC-ROC	0.996

6.3 XGBoost Performance

Metric	Value
Accuracy	0.979
Precision	0.952
Recall	0.934
F1-Score	0.943
AUC-ROC	0.992

6.4 Model Comparison

XGBoost is slightly more accurate, although both models show similar scores. Random Forest has a slightly better AUC-ROC score and curve. XGBoost also had a better recall.

6.5 Feature Importance

The following features were identified as most influential by the Random Forest model:

Rank	Feature	Importance Score
1	Age	0.322
2	Condition Count	0.140
3	Medication Count	0.138
4	Blood Pressure (Diastolic)	0.076
5	HBA1C	0.066

These top features, specifically blood pressure and age, align with real-world predictors of CVD.

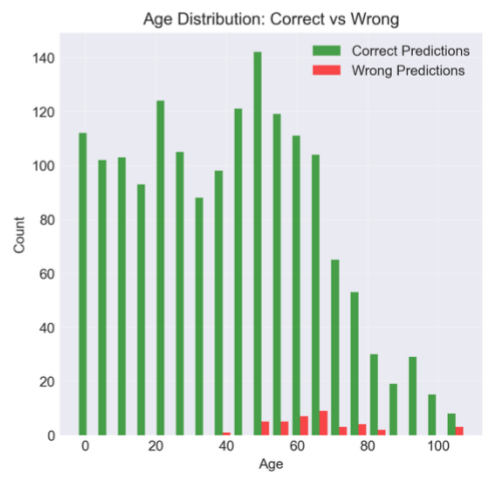
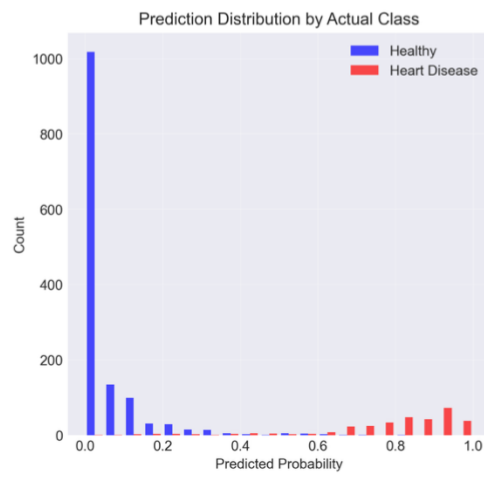
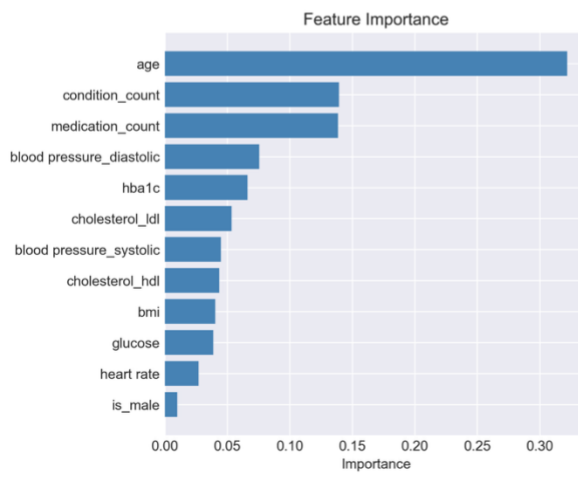
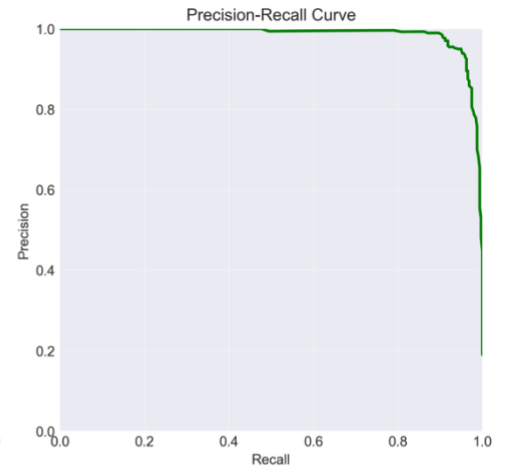
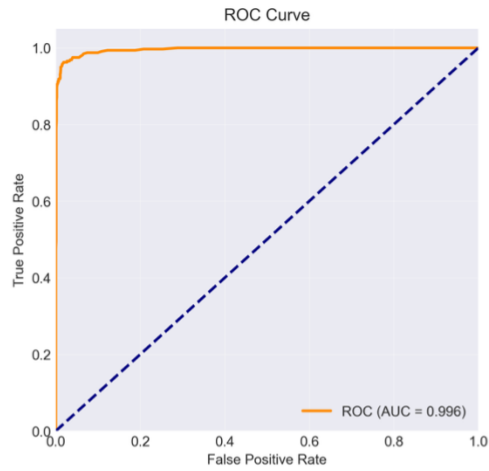
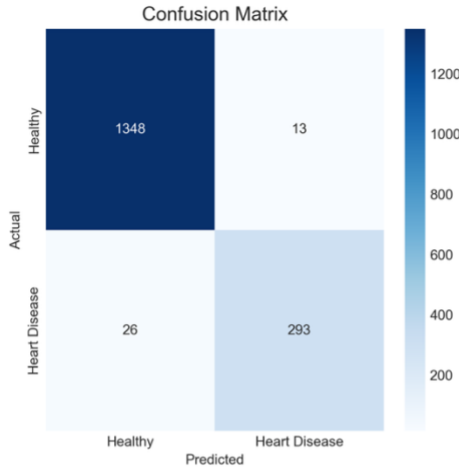
6.6 Visualizations

The following figures were generated by results.py and are included in model_results.png:

- **Figure 2:** Confusion Matrix — shows true/false positives and negatives.
- **Figure 3:** ROC Curve — plots true positive rate vs false positive rate with AUC annotation.
- **Figure 4:** Precision-Recall Curve — especially informative under class imbalance.
- **Figure 5:** Feature Importance — horizontal bar chart of top predictors.
- **Figure 6:** Prediction Probability Distribution — shows separation between predicted healthy and heart disease patients.
- **Figure 7:** Age Distribution by Prediction Correctness — shows how age relates to model errors.

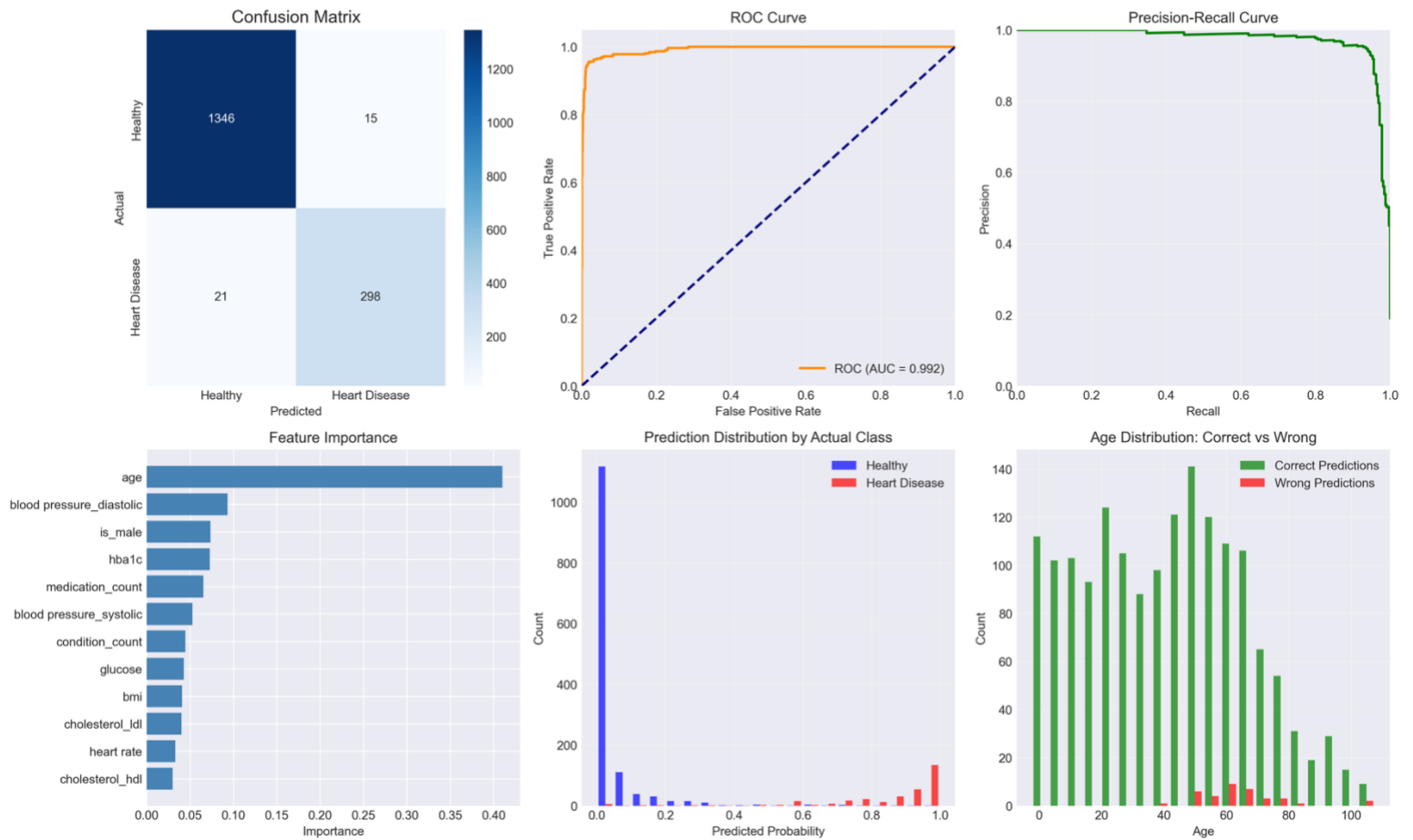
Random Forest

Heart Disease Prediction Model Results



XG Boost

Heart Disease Prediction Model Results



7. Discussion

7.1 Interpretation of Results

These classifiers had close scores in identifying patients with heart disease compared to each other, and the average model recall of 0.926 is somewhat acceptable. The models seem to perform poorly when a patient's age falls in a "gray area" for health metrics, around 40–85. With more specific, more populated, and curated data, better accuracy can be achieved. The top features also matched expected clinical risk factors like blood pressure and age.

7.2 Strengths of the Approach

- The pipeline is fully automated and reproducible from raw Synthea output.
- Features are grounded in established clinical risk factors for heart disease.
- Both SNOMED code matching and keyword-based labeling improve recall of positive cases.
- Class-balanced Random Forest handles potential label imbalance without oversampling.
- Model interpretability is maintained through feature importance analysis.

7.3 Limitations

- **Synthetic data:** Synthea-generated records may not capture the full complexity and noise of real patient data. Performance on real EHR data may differ.
- **Label definition sensitivity:** The breadth of the heart disease definition (including hypertension) affects prevalence and model performance. A stricter definition may reduce positive cases but improve label precision.
- **Missing observations:** Not all Synthea patients have every lab value recorded. Median imputation fills gaps but may introduce bias.
- **Dataset size:** The dataset was large enough for model training and testing, however the population is too small for improving model accuracy.
- **No temporal modeling:** This model uses only the latest observation per patient and does not capture the progression of disease over time.

7.4 Future Work

Several improvements could extend this project:

1. **Real patient data validation:** Testing the model on de-identified real EHR data would provide stronger evidence of clinical utility.
2. **Expanded feature set:** Including smoking status, family history, exercise level, and echocardiogram findings could improve predictive power.

3. **Temporal modeling:** Using time-series observations rather than only the latest value could capture disease progression.
 4. **Larger dataset:** Increasing the Synthea population to 5,000–10,000 patients would improve model stability and metric reliability.
 5. **Explainability:** Incorporating SHAP (SHapley Additive Explanations) values would provide patient-level explanation of predictions, which is critical for clinical trust.
 6. **Threshold optimization:** The default 0.5 classification threshold may not be optimal; calibrating the decision threshold for maximum recall could better suit clinical use.
-

8. Conclusion

This project presented an end-to-end machine learning pipeline for heart disease prediction using synthetic patient data generated by Synthea. Starting from raw clinical records, the pipeline performed patient labeling using SNOMED CT codes and keyword matching, engineering thirteen clinical features including vitals, lab values, and health complexity indicators.

Two classification models were trained and evaluated: a Random Forest classifier and an XGBoost classifier. XGBoost outperformed Random Forest by approximately 1.57% Recall. The most important predictors overall were blood pressure, age, medication/condition count, and cholesterol, which are consistent with well-established clinical risk factors for cardiovascular disease.

While the use of synthetic data limits direct clinical applicability, this work demonstrates that a structured machine learning pipeline can effectively learn to discriminate heart disease patients from a population-level EHR dataset. The methodology is reproducible, interpretable, and extensible to real data with appropriate privacy protections in place.

Full Project: <https://github.com/nmo991/Heart-Disease-Prediction>

References

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 4. World Health Organization. (n.d.). Cardiovascular diseases.
https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_
-

Appendix A: Sample Output

A.1 Sample Feature Data (heart_features.csv)

heart_features.csv

```
1 Id,age,is_male,heart_disease,patient_id,blood pressure_systolic,blood pressure_diastolic,heart rate,bmi,cholesterol_ldl,choles
2 3df79942-bcc7-d25b-bb24-d87c2f03aa63,3.660506502395619,0,0,3df79942-bcc7-d25b-bb24-d87c2f03aa63,112.0,72.0,92.0,14.8,104.9,53.
3 009f938d-fe6c-5f94-5339-497d95d98018,57.634496919917865,1,0,009f938d-fe6c-5f94-5339-497d95d98018,144.0,91.0,63.0,28.0,116.3,48.
4 b77dcfca-cb0f-d9ee-4d3a-77acdba9eaa4,1.4893908281998631,1,0,b77dcfca-cb0f-d9ee-4d3a-77acdba9eaa4,134.0,80.0,98.0,17.1,104.9,53.
5 63e732ed-d9c1-765f-f8dd-f689630869b4,41.223819301848046,1,0,63e732ed-d9c1-765f-f8dd-f689630869b4,140.0,88.0,92.0,27.7,74.5,55.
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15 94f22e77-a74c-497b-5168-f08754667cb6,13.973990417522243,1,0,94f22e77-a74c-497b-5168-f08754667cb6,139.0,83.0,66.0,3.9,104.9,53.
--
```

A.2 Model Report (model_report.txt)

Random Forest

```
=====
Dataset Size: 1680 patients
Heart Disease Prevalence: 19.0%
MODEL PERFORMANCE:
- Accuracy: 0.977
- Precision: 0.958
- Recall: 0.918
- F1-Score: 0.938
- AUC-ROC: 0.996
TOP 5 FEATURES:
- age: 0.322
- condition_count: 0.140
- medication_count: 0.138
- blood pressure_diastolic: 0.076
```

XG Boost

```
=====
Dataset Size: 1680 patients
Heart Disease Prevalence: 19.0%
MODEL PERFORMANCE:
- Accuracy: 0.979
- Precision: 0.952
- Recall: 0.934
- F1-Score: 0.943
- AUC-ROC: 0.992
TOP 5 FEATURES:
- age: 0.410
- blood pressure_diastolic: 0.093
- is_male: 0.073
- hba1c: 0.073
- medication_count: 0.065
- hba1c: 0.066
```

A.3 Python Environment

Pandas == 2.3.3

numpy == 2.3.5

scikit-learn == 1.7.2

xgboost == 3.1.2

matplotlib == 3.10.7

seaborn == 0.13.2

joblib == 1.5.2

End of Report